

Reading HP8593E Calibration Constants

Tom DG8SAQ, 24.1.2025

The GPIB command *CAL DUMP* allows to access analyzer calibration constants in unformatted numeric form. The trick is to identify which number belongs to which cal constant.

Useful information for this task is the frequency band structure of the HP8593E as shown in its data sheet:

	Band	LO harmonic = N	Center frequency
HP 8593E	0	1	9 kHz to 2.9 GHz
	1	1	2.75 to 6.5 GHz
	2	2	6.0 to 12.8 GHz
	3	3	12.4 to 19.4 GHz
	4	4	19.1 to 22 GHz
	4	4 (Option 026/027)	19.1 to 26.5 GHz

Also useful is the chapter "*Backing Up Analyzer Correction Constants*" on pages 201ff in the *8590 Series Analyzers Assembly-Level Repair* manual 08590-90316.

Helpful for identifying the CAL DUMP data is the analyzer screen obtained via menu DISPLAY CAL DATA.

8590 Series Analyzers Assembly-Level Repair manual 08590-90316 includes a detailed description of the displayed data. The relevant pages 584-587 are reproduced in the following.

DISPLAY CAL DATA Display Calibration Data

DESCRIPTION Displays the current correction-factor data generated by the **CAL FREQ** and **CAL AMPTD** self-calibration routines. Refer to [Figure 13-6](#) through [Figure 13-8](#) for an example of typical calibration data displayed when using **DISPLAY CAL DATA**. Press **PRESET** to exit the data display.

Note that the calibration data provided below is for an 8593E. The calibration data for your analyzer may be slightly different.

Figure 13-6 Typical Calibration Data: Page 1

BANDWIDTH				AMPLITUDE				NEXT PAGE
6dB	BWAMP	LC	XTL	RFATN	SGAIN	LOG	LIN	NBW
200H	-0.03	0	132	0.01	0dB	0.02		DACS
9K	0.69	128	123	0.00	10dB	0.00	-0.13	0.00
120K	-0.68	50	255	0.00	20dB	0.13	-0.40	0.00
				0.11	30dB	0.09	-0.34	0.00
3dB	BWAMP	LC	XTL	0.21	40dB	0.09	-0.47	0.00
10H	1.52	128	128	0.25	50dB	0.03		STP GAIN ZERO
30H	0.76	68	68	0.26	60dB			
100H	0.72	128	128	0.74	70dB	MCDLY		100 ANALYZER GAINS
300H	0.74	128	128					
1K	-0.06	0	223	CA	ATT	ERR	BND	GAIN
3K	-0.06	0	167	1	-0.08		0	195
10K	0.00	0	102	2	0.05		1	195
30K	-0.22	0	37	4	0.12		2	195
100K	-0.50	61	255	8	0.06		3	195
300K	0.09	111	255	16	0.00		4	195
1M	0.25	170	255					
3M	0.29	229	255	HRM	1	2	3	4
~5M	0.27	240	255	MXRB	1396	40	1549	570
								More 1
								L

Figure 13-7 Typical Calibration Data: Page 2

TUNING				
CAL	300000000	Sweepsens	<10M	0.000180161
ZERO	134090240	Sweepsens	Wide	0.000001007
FAST	54534344	Main Coil	Sens	0.059369102
MED	1076153336	FM Coilsens	Er	1.555047274
SLOW	21484809	Wide Disc	Err	4.346552849
PkOfst	-13	Wdsc	sweepsens	0.000505156
TCXO	-128	EYD	Δ slope	9.02E7
Last Cal Freq	01:41:37	SEP	08,	1992
Last Cal Amp	01:49:17	SEP	08,	1992
TRACKING GEN		FM DEMODULATION		
AOFST	0	ZERO		0
XOFST	3095	SLOPE		16981
DCORR	-32640	Narrow BW		1831
ASLOP	1.075999975	Wide BW		1808
FSLOP	0.000000120			
XSLOP	0.307900012			
				Mo

Figure 13-8 Typical Calibration Data: Page 3 (Option 130 only)

```

                                DISPL
                                CAL DA
                                -----
Narrow Bandwidth

VCXO tune 121      Gain Corr      Width Corr      Q1  Q2
Det offset 91      200 Hz 69      200 Hz 128 128
Log offset 116     10 Hz 152     10 Hz 128 128
Pole CF tune      30 Hz 76      30 Hz 68 68
P0 143           100 Hz 72     100 Hz 128 128
P1 125           300 Hz 74     300 Hz 128 128
P2 130           $G error
P3 142           0 dB 19
                   10 dB 5
                   20 dB 0
                   30 dB -18
                   40 dB 3

                                STP GA
                                ZE
                                -----
                                ANALYZ
                                GAI

                                More

                                L

```

The following text describes each column of correction data illustrated in [Figure 13-6](#), [Figure 13-7](#), and [Figure 13-8](#). The sections of [Figure 13-6](#), [Figure 13-7](#), and [Figure 13-8](#) that contain numbers retain data that is the same for all analyzers; the data is stored in memory at the factory and cannot be changed.

- **TUNING.** These miscellaneous frequency correction values correct for variations in instrument hardware performance.
- **BW-AMP.** The bandwidth amplitude corrections are mathematical offsets of the digitized video signal on the A16 processor/video assembly. Refer to the **CAL AMPTD** softkey description for more information.
- **LC-XTAL.** The A7 analog interface assembly uses these DAC values to adjust the bandwidths of the 21.4 MHz crystal and LC bandwidth filters on the A11 and A13 Bandwidth Filter assemblies. The **CAL AMPTD** self-calibration routine produces DAC values that adjust each bandwidth to within $\pm 20\%$.

The 9 kHz and 120 kHz entries are for the 6 dB EMI resolution bandwidths. All other entries are for the 3 dB resolution bandwidths.

Refer to the **CAL AMPTD** softkey description for more information.

- **RFATN.** The input attenuator corrections are mathematical offsets of the digitized video signal on the A16 processor/video assembly.

The 10 dB attenuator step is the amplitude reference used by the **CAL AMPTD** self-calibration routine while calibrating the input attenuator. Refer to the **CAL AMPTD** softkey description for more information about the calibration of the input attenuators.

Refer to “To check control of the A3 input attenuator” in [Chapter 6](#) to identify the attenuators that are in use for a given input attenuation setting.

- **SGAIN.** The step-gain and linear-gain corrections are mathematical offsets of the digitized video signal on the A16 processor/video assembly. The first six entries in the “SGAIN” column are for the 10 dB step gains on the A12 amplitude control assembly. The **CAL AMPTD** self-calibration routine uses the 10 dB step gain as an amplitude reference while calibrating the step gains.

The last four entries in the “SGAIN” column are for the 10 dB linear gains on the A14 Log Amplifier assembly.

Refer to the **CAL AMPTD** softkey description for more information about the calibration of the step-gain and linear-gain stages. Refer to [Chapter 5](#) to identify the gain stages that are in use for a given reference-level setting.

- **GAIN.** The reference-level-vernier error corrections are DAC values that calibrate the reference level at top-screen during the **CAL AMPTD** self-calibration routine.

The first five entries are fixed constants. The second five entries provide a coarse gain-adjustment of the A9 Third Converter assembly. Only one DAC value is produced by the calibration routine and is duplicated for all five entries. The A7 analog interface assembly uses the DAC value to adjust the gain of the A9 assembly. The DAC adjustment range is from 0 to 255. The higher the DAC value, the greater the gain on the A9 assembly. Refer to the **CAL AMPTD** softkey description for more information.

- **ERR.** The ERR column, next to GAIN, is the video offset “fine tuning” of the reference level calibration.

- **CAL ATT ERR.** These error corrections are mathematical offsets of the digitized video signal on the A16 processor/video assembly.

The first five entries in the “CA ATT ERR” column are correction constants that correct amplitude errors for the calibration attenuators on the A12 amplitude control assembly. Currently, the entry for the 16 dB attenuator is not used.

The calibration attenuators on the A12 amplitude control assembly are the amplitude reference for the **CAL AMPTD** self-calibration routine. The five constants are stored in nonvolatile memory initially at the factory. The constants can be retrieved from, and restored to, nonvolatile memory using the procedures for backing up analyzer correction constants in [Chapter 3](#) . When required, new constants are produced using the Cal Attenuator adjustment procedures in [Chapter 2](#) .

Refer to the **SET ATTN** softkey description for more information about the calibration attenuators on the A12 assembly.

The second five entries in the “ERR” column are one correction factor that provides a fine adjustment of the reference-level during the **CAL AMPTD** self-calibration routine. The calibration routine produces one video offset and stores it for all five entries.

The video offset is used to make a fine reference-level adjustment after a coarse reference-level adjustment is done using the DAC correction values from “RL-VENR.”

RELATED SOFTKEYS **CAL FREQ**

CAL AMPTD

DEFAULT CAL DATA

CAL STORE

CAL FETCH

SET ATTN ERROR

CAL TIMEBASE

Here are screenshots of my HP8593E's cal constants as of 11.1.12025:

Page 1

BANDWIDTH				AMPLITUDE				NEXT PAGE
6dB	BWAMP	LC	XTL	RFATN	SGAIN	LOG	LIN	NBW
200H	-0.07	0	132	-3.67	0dB	-0.04		
9K	0.88	167	103	0.00	10dB	0.00	0.37	0.00
120K	0.49	39	255	-3.77	20dB	0.37	0.20	0.00
				0.01	30dB	0.41	0.38	0.00
3dB	BWAMP	LC	XTL	-3.64	40dB	0.60	0.26	0.00
10H	1.52	128	128	0.13	50dB	0.63		
30H	1.13	128	128	-3.65	60dB			
100H	0.89	128	128	0.52	70dB			
300H	0.98	155	155					
1K	0.02	0	225					
3K	-0.04	0	167	CA	ATT	ERR	BND	GAIN
10K	-0.10	0	102	1	-0.05		0	204
30K	-0.13	0	33	2	-0.03		1	204
100K	0.46	51	255	4	0.10		2	204
300K	0.13	107	255	8	0.03		3	204
1M	-0.05	170	255	16	0.00		4	204
3M	-0.08	235	255					
~5M	-0.08	244	255	HRM	1	2	3	4
				MXRB	1600	20	1300	520

More 1

RT

Page 2

TUNING				NEXT PAGE
CAL	300000000	Sweepsens	<10M	0.000178538
ZERO	147049728	Sweepsens	Wide	0.000001013
FAST	56785714	Main	Coil Sens	0.059714206
MED	1119443312	FM	Coilsens	Er 1.528897762
SLOW	22401597	Wide	Disc Err	4.320470810
Pk0fst	19	Wdsc	sweepsens	0.000505923
TCX0	-128	EY0	Δ slope	3.562E7
Last Cal Freq	11:52:57	NOV	21, 2013	
Last Cal Amp	11:59:13	NOV	21, 2013	

ANALYZER
GAINS

More 1

GATE 1.000000000

RT

Page 3

Narrow Bandwidth				DISPLAY CAL DATA
VCX0 tune	84	Gain	Corr	
Det offset	218	200 Hz	88	
Log offset	145	10 Hz	152	
Pole CF tune		30 Hz	113	
P0	131	100 Hz	89	
P1	152	300 Hz	98	
P2	165			
P3	137			
		SG error		
		0 dB	20	
		10 dB	20	
		20 dB	0	
		30 dB	-6	
		40 dB	20	

ANALYZER
GAINS

More 1

RT

HP8593E Cal Dump:

Numbers in order of CAL DUMP output, entries identified and denoted as in screens above (page,column).

(Required) => needed for recovery after NVRAM loss and auto-cal.

CAL (2,1)

300000000

Zero (2,1)

147049728

FAST (2,1)

56785714

SLOW (2,1)

22401597

Sweepsens <10M (2,2)

1.785377E-04

Sweepsens Wide (2,2)

1.013156E-06

MED (2,1)

1119443312

FM Coilsens Er (2,2)

1.528898

GATE (2,bott)

1

Wide Disc Err (2,2)

4.320471

Main Coilsens Er (2,2)

5.971421E-02

Wdsc Sweepsens (2,2)

5.059229E-04

PkOfst (2,1)

19

MCDLY (1,R)?

140

Unknown (2 entries)

1

65

Tracking generator constants (2,1, if mounted)

AOFST

3200

DCORR

-32640

XOFST (required if TG is present)

3095

ASLOP

1.076

FSLOP

1.2E-07

XSLOP (required if TG is present)

0.3079

Unknown (5 entries)

200
200
200
200
200

BND Gain (1,5)

204
204
204
204
204

Unknown (5 entries)

130
130
130
130
130

TCXO (2,1) is -128, but this is an unsigned byte!
(required if no OCXO)

128

CA ATT ERR (1,4)

-5E-2
-3E-2
.10
3E-2
0

BND GAIN ERR (1,6)

-.10
-.10
-.10
-.10
-.10

Unknown (6 entries)

0
0
0
0
0
0

BWAMP (1,1) 200H, 9K

-7E-2
.88

BWAMP (1,1) 10H...5M

1.52

1.13
.89
.98
2E-2
-4E-2
-.10
-.13
.46
.13
-5E-2
-8E-2
-8E-2

BWAMP (1,1) 120K
.49

LC (1,2) 200H, 9K
0
167

LC (1,2) 10H...5M
128
128
128
155
0
0
0
0
51
107
170
235
244
LC (1,2) 120K
39

XTL (1,3) 200H, 9K
132
103
XTL (1,3) 200H, 9K
128
128
128
155
225
167
102
33
255
255
255
255
255

XTL (1,3) 120K
255

SGAIN LOG (1,5)
-4E-2
0
.37
.41
.60
.63

SGAIN LIN (1,6)
.37
.20
.38
.26

RFATN (1,4) (required)
-3.67
0
-3.77
1E-2
-3.64
.13
-3.65
.52

Unknown 129 entries, looks like some level correction data

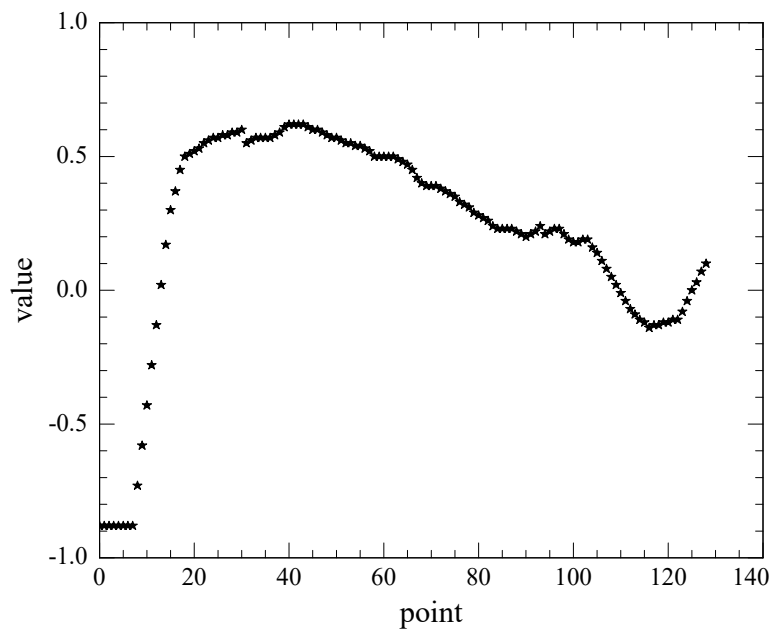
-.88
-.88
-.88
-.88
-.88
-.88
-.88
-.88
-.73
-.58
-.43
-.28
-.13
2E-2
.17
.30
.37
.45
.50
.51
.52
.53
.55
.56

.57
.57
.58
.58
.59
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.24
.21
.22
.23
.23
.21
.19
.18
.18
.19
.19
.16
.14
.11
8E-2
5E-2
2E-2
-1E-2
-4E-2
-7E-2
-9E-2
-.11
-.12
-.14
-.13
-.13
-.12
-.12
-.11
-.11
-8E-2
-4E-2
0
3E-2
7E-2

.10

Graphical depiction of above data row:



This looks like the unknown entry in the HP8591A cal dump.

Flatness Data (required)

Flatness Band 0:

Range: Start, Stop, Increment in Hz

12000000

2892000000

72000000

Flatness Data (41 entries)

-.33

-.21

-.24

-.29

1E-2

5E-2

.20

.58

.43

.69

.90

.83

1.02

.85

1.23

1.36

1.30

1.37

1.28

1.46

1.53

1.43

1.90

1.42

1.41

1.63

1.19

1.88

1.49

1.55

1.80

1.58

1.65

2.02

2.06

2.56

2.71

3.22

2.98

3.78

3.33

Flatness Band 1:

Range: Start, Stop, Increment in Hz

2750000000

6508400000

234900000

Flatness Data (17 entries)

4.52

4.14

3.89

3.48

3.24

2.88

2.82

2.85

2.52

2.72

2.33

2.41

2.64

2.26

2.56

2.62

1.98

Flatness Band 2:

Range: Start, Stop, Increment in Hz

6100000000

12724000000

184000000

Flatness Data (37 entries)

9.84

9.54

8.85

9.42

9.12

9.30

8.79

8.99

9.27

9.85

10.40

10.55

10.13

10.15

9.62

9.72

9.78

9.27

9.36

8.85

8.92

9.64

9.84
9.68
9.44
10.28
10.46
10.14
10.63
10.46
10.72
10.25
10.06
10.03
9.71
9.47
9.10

Flatness Band 3:

Range: Start, Stop, Increment in Hz

12450000000

19350000000

2300000000

Flatness Data (31 entries)

3.02
2.83
2.50
2.36
2.46
2.40
2.33
2.82
2.88
2.78
3.10
3.52
3.74
3.37
3.60
3.86
3.73
3.61
3.50
3.40
3.29
3.26
3.85
4.93
4.90
4.53
4.40
4.02
4.10

4.32
5.12

Flatness Band 4:

Range: Start, Stop, Increment in Hz

19100000000

26500000000

148000000

Flatness Data (51 entries)

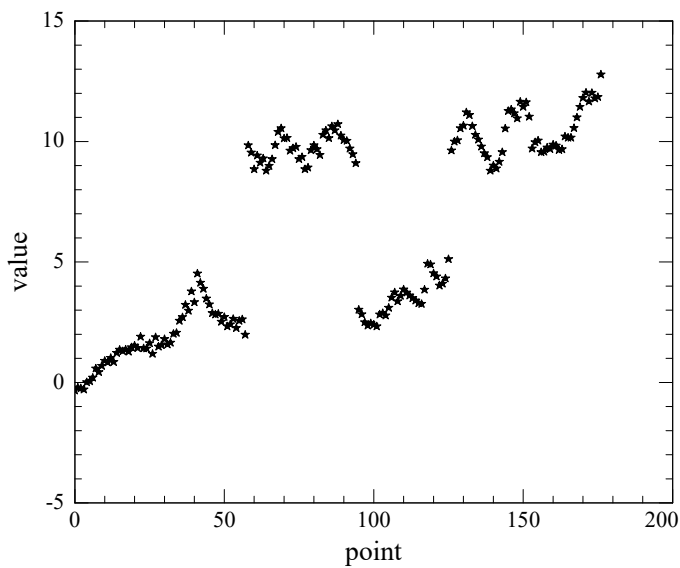
9.63
10.00
10.05
10.55
10.66
11.21
11.10
10.64
10.28
10.08
9.79
9.51
9.35
8.79
9.00
8.88
9.16
9.56
10.54
11.27
11.33
11.17
10.96
11.65
11.44
11.63
11.03
9.71
9.97
10.04
9.56
9.58
9.75
9.69
9.87
9.83
9.65
9.68
10.21
10.16
10.16
10.56
11.00

```

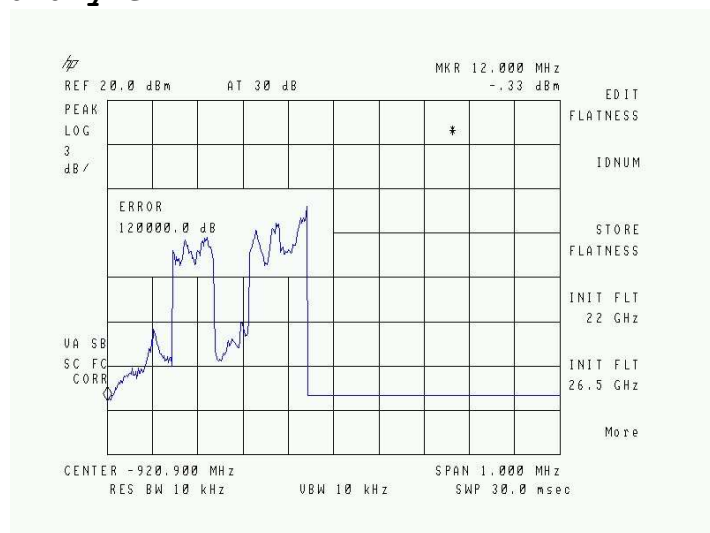
11.44
11.81
12.03
11.66
12.02
11.79
11.85
12.78
End of Flatness indicator
0

```

Graphical depiction of above data, all rows:



This indeed matches the flatness data as shown by the analyzer:



HRM MXRB (3)

```

1600
20
1300
520

```

END